ADDITIONAL NOTES FOR EACH EXPT.

MALU'S LAW

- Take intensity readings for three distinct orientations of the polariser while varying the angle of analyser by at least three full rotations.
- Take intensity readings with retarder (quarter wave plate) orientations of 0 , 45 and 90 degree
- While performing non-liner fits, make sure to put good initial parameter values and bounds of the fitting parameters. Learn to normalise intensity plots with respect to relative angle.
- Learn to Plot multiple graphs on same page

MICHAELSON INTERFEROMETER

- Learn to determine glass slide normal position from fringe shift
- Take fringe shift readings on both sides, for 10, 20, 30 , 40 ,50 fringe shifts.

SINGLE SLIT DIFFRACTION

- Take readings for two different slits and for each silt two different slit-detector distances
- Learn to normalise Intensity plots and determine slit width
- Learn to plot multiple plots on same graph

DOUBLE SLIT DIFFRACTION

- Take reading for a given slit width and slit separation
- Next, take reading for slit where slit width is same as above but slit separation is doubled
- Next, take reading from slit where slit width is doubled from that of first one but slit separation is same.
- Find slit width and separation for each of the three cases above
- Discuss the results comparing the three intensity curves with theoretical predictions

DIFFRACTION GRATING

- Take readings for 10 orders of diffraction on both sides of central maxima.
- Plot graphs of sin(θ) vs m and determine the required quantities as stated in lab manual

NEWTON'S RING

- Take readings for up to 20 diameters
- Plot a graph of D^2 vs m